```
realfora  = a12 ∈ Reals && a13 ∈ Reals  && a23 ∈ Reals  && a14 ∈ Reals  && a24 ∈ Reals &&
    a34 ∈ Reals  && a15 ∈ Reals  && a25 ∈ Reals  && a35 ∈ Reals  && a45 ∈ Reals  &&
    a16 ∈ Reals  && a26 ∈ Reals  && a36 ∈ Reals  && a46 ∈ Reals  && a56 ∈ Reals  &&
    a17 ∈ Reals  && a27 ∈ Reals  && a37 ∈ Reals  && a47 ∈ Reals  && a57 ∈ Reals  && a67 ∈ Reals ;
```

```
sx = {{0, 1}, {1, 0}};
```

```
sy = {{0, -i}, {i, 0}};
```

```
sz = {{1, 0}, {0, -1}};
```

```
I2 = {{1, 0}, {0, 1}};
```

```
g1 = sx; g2 = sy; g3 = sz;
```

```
{MatrixForm [g1], MatrixForm [g2], MatrixForm [g3]}
```

Out[ ]=
$$\left\{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right\}$$

```
g1 = KroneckerProduct [sx, g1]; g2 = KroneckerProduct [sx, g2];
g3 = KroneckerProduct [sx, g3]; g4 = KroneckerProduct [sy, IdentityMatrix [2]];
g5 = KroneckerProduct [sz, IdentityMatrix [2]];
```

```
{MatrixForm [g1], MatrixForm [g2],  MatrixForm [g3], MatrixForm [g4], MatrixForm [g5]}
```

Out[ ]=
$$\left\{\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \\ i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\right\}$$

```
I4 = IdentityMatrix [4];
```

```
Q = (g5 - I4).(g5 - I4) + g2 . g2 + g3 . g3 + g4 . g4 + g1 . g1;
```

```
MatrixForm [Q]
```

Out[ ]//MatrixForm=
$$\begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

```
v = {{1}, {0}, {0}, {0}};
```

```
MatrixForm [v]
```

Out[ ]//MatrixForm=
$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

```
First[Flatten [ConjugateTranspose  [g1 . v] . v]]
```

Out[ ]= 0

```
First[Flatten [ConjugateTranspose  [g2 . v] . v]]
```

Out[ ]= 0

```
First[Flatten [ConjugateTranspose  [g3 . v] . v]]
```

Out[ ]= 0

```
In[ ]:= First[Flatten[ConjugateTranspose [g4 . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [g5 . v] . v]]
Out[ ]= 1

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g1 . g2 - g2 . g1) . v] . v]]
Out[ ]= -2

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g1 . g3 - g3 . g1) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g1 . g4 - g4 . g1) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g1 . g5 - g5 . g1) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g2 . g3 - g3 . g2) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g2 . g4 - g4 . g2) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g2 . g5 - g5 . g2) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g3 . g4 - g4 . g3) . v] . v]]
Out[ ]= -2

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g3 . g5 - g5 . g3) . v] . v]]
Out[ ]= 0

In[ ]:= First[Flatten[ConjugateTranspose [ⅈ (g4 . g5 - g5 . g4) . v] . v]]
Out[ ]= 0
```